

Hoang Ngoc Loc

☎ 0374-830-117 | ✉ hngloc10@gmail.com | 📄 hoang-ngoc-loc | 🌐 jCool10 | 🌐 Portfolio

SUMMARY

Full-stack Software Engineer with 2+ years building and shipping production web applications end-to-end in the TypeScript ecosystem — Node.js (Express, NestJS) and React/Next.js. Have independently delivered SaaS products for international clients, including AI-integrated platforms with LangChain and OpenAI, with a focus on clean architecture (DDD, Hexagonal, microservices) and pragmatic, maintainable code.

WORK EXPERIENCE

Full-stack Developer

24H.DEV CO., LTD

Jul 2024 – Present

Da Nang, Vietnam

- Sole Node.js + React/Next.js developer on multiple outsourced products for international clients — owning architecture, backend, and frontend end-to-end, from requirements to production deployment.
- Built an AI-powered SaaS compliance platform (FATCA/CRS) for a European fintech client (under NDA): Next.js + React, GraphQL (Apollo Server) + REST APIs, Prisma + PostgreSQL, NextAuth + Azure AD SSO; integrated LangChain/LangGraph with OpenAI and Azure OpenAI to build streaming entity-classification workflows that automate FATCA/CRS classification of legal entities.
- Contributed to a large-scale recruitment & job-matching platform for an international HR-tech client across 18+ Node.js microservices (BFF, profile, jobs, messaging, CV-parser, matches-scheduler) and React frontends; built and improved features across the candidate-search pipeline and profile data-integrity flows.

Freelance Front-end Developer

MLOpsVN

Mar 2023 – Sep 2023

Remote

- Built the official website for MLOps Marathon 2023 — the first MLOps competition in Vietnam — with Next.js + SCSS Modules, fully mobile-responsive and performance-optimized.
- Deployed and managed the production site on Vercel — <https://mlops.vn/>.

PERSONAL PROJECTS

jCommerce — Microservices E-Commerce Platform

2025 – Present

NestJS, Next.js, TypeScript, PostgreSQL, RabbitMQ, Elasticsearch, Redis, Stripe, Docker, Turborepo

- Self-driven side project to explore production microservices: 6 NestJS services (API Gateway, Auth, Catalog, Search, Order, Email Worker) and 2 Next.js apps (Storefront, Admin) in a Turborepo monorepo.
- Applied real-world patterns end-to-end: Hexagonal Architecture, event-driven flows via RabbitMQ with the Transactional Outbox, Saga orchestration with compensation for the checkout flow, and idempotent Stripe webhook handling.
- Shipped with the full operational story: blue-green Elasticsearch indexing, BullMQ email pipeline, JWT auth, and a complete observability stack (OpenTelemetry, Grafana, Loki, Tempo, Prometheus, Sentry).
- Github: <https://github.com/jCool10/jCommerce>

AnythingRAG - RAG Chat with anything

2025

Next.js, Express.js, TypeScript, Python (FastAPI), LangChain, Milvus, MongoDB, AWS S3, Docker

- A multi-user RAG application that lets users upload PDFs and chat with them in natural language, returning source-attributed answers across English and Vietnamese.
- Built as three containerized services — Next.js client, Node.js / Express API, and a Python FastAPI RAG service — with a hybrid retrieval pipeline (Milvus dense embeddings + BM25 sparse, fused via RRF) and cross-encoder reranking.
- Secured with JWT (RS256) auth and per-user RSA keypair encryption for workspace isolation; persistence on MongoDB + AWS S3, all orchestrated via Docker Compose.
- Github: <https://github.com/jCool10/capstone-project>

EDUCATION

Da Nang University of Science and Technology
Bachelor of Computer Engineering

Da Nang, Vietnam
Oct 2020 – Sep 2025

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++

Back-end: Node.js, Express.js, NestJS, REST APIs, GraphQL (Apollo), Socket.IO, Microservices, Event-driven (RabbitMQ, Kafka), DDD / Hexagonal

Front-end: React.js, Next.js (App Router), Redux, React Query, React Hook Form, Tailwind CSS, Sass/SCSS, shadcn/ui, Material UI

Databases & Search: PostgreSQL, MySQL, MongoDB, Prisma, Redis, Elasticsearch, Milvus, pgvector

Cloud & DevOps: AWS (S3, EC2), Azure (Blob, AD, OpenAI, Document Intelligence), GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Vercel, Nginx

AI / ML: LangChain, LangGraph, OpenAI API, RAG, HuggingFace embeddings, Pinecone, Milvus

Observability & Tooling: OpenTelemetry, Grafana, Loki, Tempo, Prometheus, Sentry, Postman, Swagger

Methodology: Agile / Scrum, Git / GitHub, Code Review